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Electronic device

Abstract

An electronic device is provided. The electronic device includes a substrate, a metal layer, a plurality of diodes, a first transistor, and a second transistor. The metal layer is overlapped with the substrate and includes a plurality of holes. The diodes are disposed on the substrate. The first transistor and the second transistor are disposed on the substrate and electrically connected to one of the diodes, the gate electrodes of the first transistor and the second transistor overlap a same semiconductor, and the one of the diodes overlaps the first transistor and doesn't overlap the second transistor.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of application Ser. No. 18/149,380, filed Jan. 3, 2023 now U.S. Pat. No. 12,067,185), which is a Continuation of application Ser. No. 17/752,945, filed May 25, 2022 (now U.S. Pat. No. 11,556,203), which is a Continuation of application Ser. No. 17/130,172, filed Dec. 22, 2020 (now U.S. Pat. No. 11,366,545), which is a Continuation of application Ser. No. 16/554,685, filed Aug. 29, 2019, which is a Continuation of application Ser. No. 15/670,120, filed Aug. 7, 2017, which claims the benefit of provisional Application No. 62/500,539, filed May 3, 2017, the entirety of which is incorporated by reference herein.

BACKGROUND

Technical Field

(1) The disclosure relates to display devices, and in particular to display devices including light-emitting diodes.

Description of the Related Art

(2) As digital technology develops, display devices are becoming more widely used in our society. For example, display devices have been applied in modern information and communication devices such as televisions, notebook computers, desktop computers, mobile phones (e.g., smartphones). In addition, each generation of display devices has been developed to be thinner, lighter, smaller, and more fashionable.

(3) Among the various types of display devices, light-emitting diode (LED) display devices are becoming popular due to the advantages of LEDs which include high efficiency and long lifespans.

(4) However, existing LED display devices have not been satisfactory in every respect. For example, the conductive line or conductive electrode may be easily cracked during the formation of a curved display device).

BRIEF SUMMARY

(5) Some embodiments of the disclosure provide a display device. The display device includes a substrate having a first edge and a second edge opposite to the first edge, wherein there is a first distance between the first edge and the second edge; a plurality of light-emitting units disposed on the substrate; and a conductive line disposed on the substrate and electrically connected to at least one of the plurality of light-emitting units, wherein the conductive line includes an undulating edge, wherein the conductive line has an extending length, and wherein the extending length is greater than or equal to half of the first distance and is less than or equal to the first distance.

(6) Some embodiments of the disclosure provide a display device. The display device includes a substrate; a plurality of light-emitting units disposed on the substrate; and a conductive line disposed on the substrate and electrically connected to at least one of the plurality of light-emitting units, wherein the conductive line includes a plurality of openings.

(7) A detailed description is given in the following embodiments with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure can be more fully understood from the following detailed description when read with the accompanying figures. It is worth noting that in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1A is a schematic side view of the curved display device **10** according to some embodiments of the present disclosure.

(3) FIG. 1B is a schematic side view of the curved display device **10'** according to some

embodiments of the present disclosure.

(4) FIG. 2A is a schematic top view of the display device **20** according to First Embodiment of the present disclosure.

(5) FIG. 2B is a schematic top view of a portion of the display region **202** of the display device **20** according to some embodiments of the present disclosure.

(6) FIG. 2C is a schematic top view of a portion of the display region **202** of the display device **20** according to some embodiments of the present disclosure.

(7) FIG. 2D is a schematic top view of a portion of the display region **202** of the display device **20** according to some embodiments of the present disclosure.

(8) FIG. 2E is a schematic top view of a portion of the display region **202** of the display device **20** according to some embodiments of the present disclosure.

(9) FIG. 2F is a schematic top view of a portion of the conductive line **208** according to some embodiments of the present disclosure.

(10) FIG. 2G is a schematic top view of a portion of the conductive line **208** according to some embodiments of the present disclosure.

(11) FIG. 2H is a schematic top view of the display device **20** according to some embodiments of the present disclosure.

(12) FIG. 2I is a schematic top view of the display device **20** according to some embodiments of the present disclosure.

(13) FIG. 2J is a schematic top view of the display device **20** according to some embodiments of the present disclosure.

(14) FIG. 3A is a schematic top view of a portion of the display region **202** of the display device **30** according to Second Embodiment of the present disclosure.

(15) FIG. 3B is a schematic top view of a portion of the display region **202** of the display device **30** according to some embodiments of the present disclosure.

(16) FIG. 3C is a schematic top view of a portion of the display region **202** of the display device **30** according to some embodiments of the present disclosure.

(17) FIG. 3D is a schematic top view of a portion of the conductive electrode **210** according to some embodiments of the present disclosure.

(18) FIG. 3E is a schematic top view of a portion of the conductive electrode **210** according to some embodiments of the present disclosure.

(19) FIG. 4A is a schematic cross-sectional view of the display device **40** according to some embodiments of the present disclosure.

(20) FIG. 4B is a schematic cross-sectional view of the display device **40** according to some embodiments of the present disclosure.

(21) FIG. 4C is a schematic cross-sectional view of the display device **40** according to some embodiments of the present disclosure.

(22) FIG. 5 is a schematic cross-sectional view of the display device **50** according to Third Embodiment of the present disclosure.

(23) FIG. 6A is a schematic side view of the display device **60** according to Fourth Embodiment of the present disclosure.

(24) FIG. 6B is a schematic side view of the display device **70** according to an embodiment of the present disclosure.

(25) FIG. 7A is a schematic top view of the display device **80** according to Fifth Embodiment of the present disclosure.

(26) FIG. 7B is a schematic top view of the display device **80** according to some embodiments of the present disclosure.

(27) FIG. 7C is a schematic top view of the display device **80** according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(28) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various embodiments. This repetition is for simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(29) Some embodiments are described below. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like features. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

(30) FIG. 1A is a schematic side view of the curved display device **10** according to some embodiments of the present disclosure. As shown in FIG. 1A, the curved display device **10** includes one or more curved portions **100**. For example, the curved portions **100** can include the end portions of the display device **10**. In some embodiments, the curved portions **100** (e.g., two curved portions **100**) are symmetrical with respect to the center line C of the curved display device **10** as shown in FIG. 1A.

(31) FIG. 1B is a schematic side view of another curved display device **10'** according to embodiments of the present disclosure. As shown in FIG. 1B, the curved display device **10'** can have curved portion **100'** including the center portion of the display device **10'**.

(32) It should be noted that the curved display devices (e.g., curved display device **10** or curved display device **10'**) of some embodiments of the present disclosure can be foldable display devices. However, in other embodiments, the shape of the curved display device is substantially fixed.

First Embodiment

(33) First Embodiment of the present disclosure provides some display devices which can be used to form curved display devices. The conductive lines of the display devices of First Embodiment may have a wavy shape and/or include one or more openings, which may avoid or reduce the formation of cracks in the step of using these display devices to form curved display devices.

(34) FIG. 2A is a schematic top view of the display device **20** according to First Embodiment of the present disclosure. The display device **20** can be used to form a curved display device (e.g., a curved display device with a side view the same as, or similar to the curved display devices **10** or **10'** discussed above). For example, the display device **20** can be bent to form a curved display device.

(35) As shown in FIG. 2A, the display device **20** can include a display region **202** and a non-display region **204** adjacent to the display region **202**. For example, the non-display region **204** can surround or enclose the display region **202**, as shown in FIG. 2A. The display region **202** can refer to the display region in the display device **20** in which the light-emitting units (e.g., light-emitting diodes) are disposed. On the other hand, the non-display region **204** refers to a region other than the display region **202** in the display device **20**. For example, an integrated circuit (IC), a demultiplexer (DEMUX), a gate driver on panel (GOP), other applicable devices or components, or a combination thereof may be disposed in the non-display region **204**. For example, the devices or components disposed in the non-display region **204** may be electrically connected to the light-emitting units (e.g., light-emitting diodes) disposed in the display region **202** to provide some desired functions (e.g., providing driving signal to the light-emitting units disposed in the display region **202**).

(36) As shown in FIG. 2A, the display device **20** includes a substrate **200** having a first edge **200a**

and a second edge **200b** opposite to the first edge **200a**. In some embodiments, the first edge **200a** and the second edge **200b** are substantially parallel to each other and are spaced apart by a first distance $D_{sub.1}$ (e.g., 5 cm to 10 cm). For example, the substrate **200** of the display device **20** may include polyimide, glass, other applicable materials, or a combination thereof. It should be noted that the elements of the display device **20** formed on the substrate **200** are not shown in FIG. 2A, and will be discussed in the following paragraph.

(37) Then, referring to FIG. 2B, a schematic top view of a portion of the display region **202** of the display device **20** is illustrated. As shown in FIG. 2B, the display region **202** can include a plurality of light-emitting units **206** (e.g., light-emitting diodes), one or more conductive lines **208**, and one or more conductive electrodes **210**.

(38) In this embodiment, a sub-pixel **212** has two light-emitting units **206** (e.g., light-emitting diodes). However, the sub-pixel **212** may have less or more than two light-emitting units **206** with the same color in other embodiments. In some embodiments, adjacent sub-pixels **212** (e.g., three sub-pixels **212**) can form a pixel. For example, the pixel can include sub-pixels **212** containing light-emitting units **206** of different colors (e.g., red, green, blue, or a combination thereof).

(39) For example, the material of the conductive electrodes **210** can include, but is not limited to, indium tin oxide (ITO), tin oxide (SnO), indium zinc oxide (IZO), indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO), antimony tin oxide (ATO), antimony zinc oxide (AZO), metal, or a combination thereof. For example, the material of the conductive lines **208** can include, but is not limited to, indium tin oxide (ITO), tin oxide (SnO), indium zinc oxide (IZO), indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO), antimony tin oxide (ATO), antimony zinc oxide (AZO), metal, or a combination thereof. For example, the conductive electrodes **210** and the conductive lines **208** may be respectively formed by a deposition process followed by a lithography process and an etching process.

(40) In some embodiments, the conductive line **208** can be a common line, the conductive electrode **210** can be a common electrode, and the conductive line **208** can be electrically connected to at least one of the light-emitting units **206** through the conductive electrode **210**.

(41) In some embodiments, the one or more conductive electrodes **210** can be disposed on the one or more conductive lines **208** and the plurality of light-emitting units **206**. In other words, the one or more conductive lines **208** and the plurality of light-emitting units **206** can be disposed between the substrate **200** and the one or more conductive electrodes **210**.

(42) In some embodiments, the conductive line **208** may have a wavy shape. For example, as shown in FIG. 2B, the conductive line **208** may include an undulating edge **208a** (e.g., an undulating edge extending along a direction substantially perpendicular to the first edge **200a** and/or the second edge **200b** of the substrate **200**), and thus the formation of cracks in the conductive line **208** during the step of forming a curved display device (e.g., bending the display device **20** to form a curved display device) may be avoided or reduced.

(43) In some embodiments, as shown in FIG. 2B, the conductive line **208** can extend along a direction substantially perpendicular to the first edge **200a** and/or the second edge **200b** of the substrate **200** (referring to FIG. 2A), and an extending length $L_{sub.1}$ (e.g., a distance measured in a direction perpendicular to the first edge **200a** and/or the second edge **200b** of the substrate **200**) of the conductive line **208** can be greater than or equal to half of the first distance $D_{sub.1}$ and be less than or equal to the first distance $D_{sub.1}$ (i.e., $0.5D_{sub.1} \leq L_{sub.1} \leq D_{sub.1}$). In some embodiments, since the extending length $L_{sub.1}$ is greater than or equal to half of the first distance $D_{sub.1}$ and less than or equal to the first distance $D_{sub.1}$, a better device performance can be obtained.

(44) FIGS. 2C, 2D, 2E, 2F, and 2G illustrate some conductive lines **208** of some embodiments of the present disclosure which also may avoid or reduce the formation of cracks during the step of forming a curved display device.

(45) Referring to FIG. 2C, one difference between the embodiments illustrated in FIG. 2C and FIG.

2B is that the conductive line **208** of the embodiments illustrated in FIG. 2C further includes one or more openings **214**. For example, a patterning process including a lithography process followed by an etching process may be used to form the one or more openings **214** in the conductive line **208**. In some embodiments, the opening **214** may have a shape including a first curved portion **214a** (e.g., the round corners of the rectangular opening **214**), and thus the mechanical property of the conductive line **208** may be further improved. In other embodiments, the opening **214** may also be substantially oval-shaped (e.g., as shown in FIG. 2D), substantially square, substantially rectangular, substantially round, substantially oblong, substantially triangular, polygon, irregularly-shaped, other applicable shapes, or a combination thereof.

(46) Referring to FIG. 2E, one difference between the embodiments illustrated in FIG. 2C and FIG. 2E is that the conductive line **208** of the embodiments illustrated in FIG. 2E includes a substantially straight edge **208a** (e.g., a straight edge extending along a direction substantially perpendicular to the first edge **200a** and/or the second edge **200b** of the substrate **200**). Notably, the opening **214** of the embodiments illustrated in FIG. 2E may also be substantially oval-shaped, substantially square, substantially rectangular, substantially round, substantially oblong, substantially triangular, polygon, irregularly-shaped, other applicable shapes, or a combination thereof.

(47) In some embodiments, as shown in FIGS. 2C, 2D, and 2E, the one or more openings **214** may be located between adjacent conductive electrodes **210**.

(48) In some embodiments, as shown in FIG. 2F, a single conductive line **208** may include a plurality of openings **214** having different shapes or sizes to increase design flexibility.

(49) In some embodiments, as shown in FIG. 2G, the conductive line **208** may have a partially undulating edge **208a**. For example, the edge **208a** of the conductive line **208** may include a substantially straight portion **208a'** and a wavy portion **208a''**, where the wavy portion is provided at weak points that are prone to cracking.

(50) In some embodiments, the conductive lines **208** are common lines that can be used to transmit common signals from an IC disposed in the non-display region **204** of the display device **20** to the light-emitting units **206** disposed in the display region **202** of the display device **20**. For example, as shown in FIG. 2H, the common signals can be transmitted from an IC **216** into the display region **202** of the display device **20** through the common lines **208** in a direction substantially perpendicular to the first edge **200a** and/or the second edge **200b** of the substrate **200**. As shown in FIG. 2H, the common lines **208** may be electrically connected to the IC **216** through another common line **218** disposed in the non-display region **204** of the substrate **200**. In some embodiments, the common line **218** may also have a wavy shape, and/or include a plurality of openings as do the common lines **208** discussed above. In some embodiments, the common line **218** and the common lines **208** may be disposed in different layers, and in such cases one or more vias may be used to electrically connect the common line **218** to the common lines **208**. In some embodiments, the common line **218** and the common lines **208** may be disposed in the same layer, and the common line **218** and the common lines **208** may be formed by a same process (e.g., deposition, lithography, etching, other applicable processes, or a combination thereof). As shown in FIG. 2H, a demultiplexer (DEMUX) **222'** electrically connected to the IC **216**, and a gate driver on panel **222** electrically connected to the IC **216** may also be disposed in the non-display region **204** of the display device **20**. In addition, as shown in FIG. 2H, the IC **216** may be electrically connected to a printed circuit board **220**.

(51) In some embodiments, as shown in FIG. 2I, the common signals can be transmitted from the IC **216** into the display region **202** of the display device **20** through common lines **224** extending along a direction substantially parallel to the first edge **200a** and/or the second edge **200b** of the substrate **200**, and thus the common signals are transmitted into the display region **202** of the display device **20** in the direction substantially parallel to the first edge **200a** and/or the second edge **200b**. In some embodiments, the common lines **224** may also have a wavy shape, and/or include a plurality of openings as do the common lines **208** discussed above.

(52) In some embodiments, as shown in FIG. 2J, the common signals can be transmitted into the display region **202** of the display device **20** through both the common lines **208** extending along a direction that is substantially perpendicular to the first edge **200a** and/or the second edge **200b** of the substrate **200**, and the common lines **224** extending along a direction that is substantially parallel to the first edge **200a** and/or the second edge **200b** of the substrate **200**. In other words, the common signals can be transmitted into the display region **202** of the display device **20** in two directions substantially perpendicular to each other.

(53) Understandably, FIGS. 2H, 2I, and 2J are schematic top views of some display devices according to some embodiments of the present disclosure, and the elements illustrated in these FIGS. may be disposed in the same layer or in different layers.

(54) It should be noted that the curved display devices formed of the display devices of the above embodiments may include technical features which are the same as or similar to the technical features of the display devices of the above embodiments. For example, the conductive lines may still have a wavy shape and/or include one or more openings in the curved display devices formed of the display devices of the above embodiments. These curved display devices are intended to be included within the scope of the present disclosure.

Second Embodiment

(55) FIG. 3A is a schematic top view of a portion of the display region **202** of the display device **30** according to Second Embodiment of the present disclosure. One difference between Second Embodiment and the above embodiments is that the one or more conductive electrodes **210** (e.g., a common electrode) may have a wavy shape and/or include one or more openings, which may avoid or reduce the formation of cracks in the one or more conductive electrodes during the step of forming a curved display device (e.g., bending the display device **30** to form a curved display device).

(56) As shown in FIG. 3A, the conductive electrode **210** may have a wavy shape. For example, as shown in FIG. 3A, the conductive electrode **210** may include an undulating edge **210a** (e.g., an undulating edge extending along a direction substantially parallel to the first edge **200a** and/or the second edge **200b** of the substrate **200**).

(57) In some embodiments, as shown in FIG. 3B, the conductive electrode **210** may have one or more openings **302**. For example, a patterning process including a lithography process followed by an etching process may be used to form the one or more openings **302** in the conductive electrode **210**. In some embodiments, the opening **302** may have a shape including a first curved portion **302a** (e.g., the round corners of the rectangular opening **302**), and thus the mechanical property of the conductive electrode **210** may be further improved. In other embodiments, the opening **302** may also be substantially oval-shaped, substantially square, substantially rectangular, substantially round, substantially oblong, substantially triangular, polygon, irregularly-shaped, other applicable shape, or a combination thereof.

(58) In some embodiments, as shown in FIG. 3B, the area of one of the openings **302** is smaller than the area of one of the light-emitting units **206** (e.g., the area of the bottom surface or top surface of one of the light-emitting units **206**), and thus the mechanical property of the conductive electrode **210** may be further improved.

(59) In some embodiments, as shown in FIG. 3C, a single bulk conductive electrode **210** including a plurality of openings **302** can be provided. In addition, the single conductive electrode **210** may cover a plurality of sub-pixels **212**.

(60) In some embodiments, as shown in FIGS. 3A, 3B, and 3C, the light-emitting units **206** are fully covered by the conductive electrodes **210**.

(61) FIGS. 3D and 3E illustrate some conductive electrodes **210** according to some embodiments of the present disclosure which also may avoid or reduce the formation of cracks during the step of forming a curved display device.

(62) In some embodiments, as shown in FIG. 3D, a single conductive electrode **210** may include a

plurality of openings **302** having different shapes or sizes to increase design flexibility.

(63) In some embodiments, as shown in FIG. 3E, the conductive electrode **210** may have a partially undulating edge **210a**. For example, the edge **210a** of the conductive electrode **210** may include a substantially straight portion **210a'** and a wavy portion **210a''**, where the wavy portion is provided at weak points that are prone to cracking.

(64) In some other embodiments, the design of the conductive electrodes **210** (e.g., the wavy shape, and/or the opening therein) discussed above can be applied to a common electrode, a touch electrode, other conductive electrodes, or a combination thereof of a display device. The details will be discussed in the following paragraph.

(65) FIG. 4A is a schematic cross-sectional of the display device **40** according to some embodiments of the present disclosure. The display device **40** can be used to form a curved display device (e.g., a curved display device with a side view the same as, or similar to the curved display devices **10** or **10'** discussed above). For example, the display device **40** can be bent to form a curved display device.

(66) As shown in FIG. 4A, the display device **40** includes a substrate **400**. For example, the substrate **400** can be made of glass, polyimide, other applicable materials, or a combination thereof. A buffer layer **402** may be optionally formed on the substrate **400** to reduce the lattice mismatch. For example, the buffer layer **402** can be made of silicon oxide, silicon nitride, other applicable materials, or a combination thereof. A semiconductor layer **404** may be formed on the buffer layer **402**. For example, the semiconductor layer **404** may include one or more doped regions **404a** which may serve as source/drain regions. A gate insulator **406** may be formed on the semiconductor layer **404**. For example, the gate insulator **406** may be made of silicon oxide, other applicable materials, or a combination thereof. One or more gate electrodes **408** may be formed on the gate insulator **406**, and a first insulating layer **410** (e.g., silicon oxide) may be formed on the one or more gate electrodes **408**. A second insulating layer **412** may be formed on the first insulating layer **410**. For example, the second insulating layer **412** may be made of a material (e.g., silicon nitride) that is different from the material of the first insulating layer **410**. A third insulating layer **414** may be formed on the second insulating layer **412**. For example, the third insulating layer **414** may be made of silicon nitride, organic materials (e.g., Acrylic based organic material), other applicable insulating materials, or a combination thereof. A pixel defining layer (PDL) **416** may be formed on the third insulating layer **414**. For example, the pixel defining layer **416** (e.g. an insulating layer) may be made of photosensitive polyimide, acrylic, siloxane, novolac materials, other applicable materials, or a combination thereof. A reflector **420** may be formed on the pixel defining layer **416**. As shown in FIG. 4A, the reflector **420** may extend from a top surface of the pixel defining layer **416** into the pixel defining layer **416** and the third insulating layer **414**. For example, the reflector **420** may be made of metal (e.g., Al or Ag), metal alloys (e.g., metal alloys of Al or Ag), other applicable materials, or a combination thereof. One or more light-emitting units **424** (e.g., light-emitting diodes) may be formed on the reflector **420**. As shown in FIG. 4A, a filling material may be formed on the reflector **420** and around the one or more light-emitting units **424**. For example, the filling material **422** may be made of organic material, anisotropic conductive film (ACF) bonding material, other applicable materials, or a combination thereof. In some embodiments, the filling material can include quantum dots therein. As shown in FIG. 4A, one or more conductive lines **418** may be formed on the pixel defining layer **416**. For example, the conductive lines **418** may also have a wavy shape and/or include one or more openings as do the conductive lines **208** discussed above. As shown in FIG. 4A, the display device **40** may also include a fourth insulating layer **426** formed on the conductive lines **418**, the light-emitting units **424**, and the pixel defining layer **416**. For example, the fourth insulating layer **426** may be made of an organic material, inorganic/organic stacked layer (e.g., SiNx/organic/SiNx stacked layer), other applicable insulating materials, or a combination thereof. As shown in FIG. 4A, a conductive electrode **428** which may be configured to electrically connect the conductive line **418** to the light-emitting unit **424** may be

formed on the conductive lines **418**, the light-emitting units **424**, and the fourth insulating layer **426**.

(67) As shown in FIG. **4A**, one or more conductive structures **434** may be formed in the display device **40** to electrically connect different elements of the display device **40**.

(68) In some embodiments, the conductive electrode **428** may include a common electrode, a touch electrode, or a combination thereof. As noted above, the design of the conductive electrodes **210** (e.g., the wavy shape, and/or the opening therein) discussed above can be applied to the conductive electrode **428** (e.g., a common electrode, a touch electrode, or a combination thereof), such that the formation of cracks during the step of forming a curved display device may be avoided or reduced.

(69) In some embodiments, a single bulk conductive electrode **428** may simultaneously serve as a common electrode and a touch electrode, and thus the route for transmitting common signals and the route for transmitting touch signals may be the same. However, in other embodiments, as shown in FIG. **4B**, two conductive electrodes **428a** and **428b** may respectively serve as a touch electrode and a common electrode, and thus the route for transmitting touch signals and the route for transmitting common signals may be different.

(70) In some embodiments, the touch electrode and the common electrode may be formed in different layers. For example, as shown in FIG. **4C**, one or more touch electrodes **432** can be formed on the common electrode **428**, and an insulating layer **430** may be formed between the one or more touch electrodes **432** and the common electrode **428**. In some embodiments, as shown in FIG. **4C**, the one or more touch electrodes **432** and the light-emitting units **424** do not overlap, and thus a better device performance may be obtained.

(71) Notably, more layers of conductive electrodes can be formed on the conductive electrode **428** according to, for example, the application of the display device. For example, more layers of touch electrodes can be formed on the common electrode **428** and the touch electrodes **432** of the display device illustrated in FIG. **4C**.

(72) In some embodiments, as shown in FIG. **4A**, the insulating layer under the light-emitting units **424** (e.g., insulating layers **414**) may include a recess or a trench **414a** therein, which may further reduce the formation of cracks in the display device during the step of forming a curved display device.

(73) It should be noted that the curved display devices formed of the display devices of the above embodiments may include technical features that are the same as or similar to the technical features of the display devices of the above embodiments. For example, the conductive electrodes may still have a wavy shape and/or include one or more openings in the curved display devices formed of the display devices of the above embodiments. These curved display devices are intended to be included within the scope of the present disclosure.

Third Embodiment

(74) FIG. **5** is a schematic cross-sectional of the display device **50** according to Third Embodiment of the present disclosure. The display device **50** can be used to form a curved display device (e.g., a curved display device with a side view the same as, or similar to the curved display devices **10** or **10'** discussed above). For example, the display device **50** can be bent to form a curved display device.

(75) The display device **50** may be similar to the display devices of the above embodiments (e.g., the display device **40**), and the difference between the display device **50** and the display devices of the above embodiments will be discussed in the following paragraph.

(76) As shown in FIG. **5**, the pixel defining layer **416** may include a first portion **416a** overlapping with the conductive line **418**, and a second portion **416b** not overlapping with the conductive line **418**. As shown in FIG. **5**, the second portion **416b** of the pixel defining layer **416** is adjacent to the first portion **416a** of the pixel defining layer **416** (e.g. an insulating layer). In some embodiments, a thickness $T_{sub.1}$ of the first portion **416a** can be greater than a thickness $T_{sub.2}$ of the second portion **416b** (e.g., a minimum thickness of the second portion **416b**), which may improve the

mechanical property of the display device **50** (e.g., avoiding or reducing the formation of cracks in the display device **50** during the step of bending the display device **50** to form a curved display device).

(77) As shown in FIG. 5, the fourth insulating layer **426** may include a first portion **426a** overlapping with the conductive electrode **428**, and a second portion **426b** not overlapping with the conductive electrode **428**. As shown in FIG. 5, the second portion **426b** of the fourth insulating layer **426** is adjacent to the first portion **426a** of the fourth insulating layer **426**. In some embodiments, a thickness $T_{sub.4}$ of the second portion **426b** can be greater than a thickness $T_{sub.3}$ of the first portion **426a** (e.g., a minimum thickness of the first portion **426a**), which may improve the mechanical property of the display device **50** (e.g., avoiding or reducing the formation of cracks in the display device **50** during the step of bending the display device **50** to form a curved display device).

(78) It should be noted that the curved display devices formed of the display devices of the above embodiments may include technical features which are the same as or similar to the technical features of the display devices of the above embodiments (e.g., the technical features of the insulating layers discussed in paragraphs [0086] and [0087]). These curved display devices are intended to be included within the scope of the present disclosure.

Fourth Embodiment

(79) Fourth Embodiment of the present disclosure also provides some display devices which may avoid or reduce the formation of cracks in the step of using these display devices to form curved display devices.

(80) Referring to FIG. 6A, a schematic side view of the display device **60** according to Fourth Embodiment of the present disclosure is provided. Although only the substrate **200** and the light-emitting units (e.g., light-emitting units) **206a** and **206b** are shown in FIG. 6A for the interest of simplicity, one skilled in the art can understand that additional layers and/or components discussed in the above embodiments may also be disposed in the display device **60**. For example, some insulating layers may be formed between the substrate **200** and the light-emitting units (**206a** and **206b**) of the display device **60**, and one or more conductive electrodes may be formed on the light-emitting units **206a** and **206b**.

(81) For example, the display device **60** can be bent to form a curved display device (e.g., curved display device **10'** shown in FIG. 1B). As shown in FIG. 6A, the display device **60** can include a portion **60a** and a portion **60b**. For example, the portion **60a** can correspond to a curved portion of a curved display device (e.g., the curved portion **100'** of the curved display device **10'** illustrated in FIG. 1B) formed of the display device **60**, and the portion **60b** can correspond to a non-curved portion of the curved display device formed of the display device **60**.

(82) In some embodiments, the dimension of one of the light-emitting units **206a** in the portion **60a** may be smaller than the dimension of one of the light-emitting units **206b** in the portion **60b**, which may avoid or reduce the formation of cracks in the step of using the display device **60** to form a curved display device. For example, the bottom surface area of one of the light-emitting units **206a** may be smaller than the bottom surface area of one of the light-emitting units **206b**. In some embodiments, the ratio of the bottom surface area of one of the light-emitting units **206a** to the bottom surface area of one of the light-emitting units **206b** may be 0.3 to 0.95.

(83) In some embodiments, as shown in FIG. 6A, the density (e.g., number of light-emitting units per unit area) of the light-emitting units **206a** in the portion **60a** may be smaller than the density of the light-emitting units **206b** in the portion **60b**, which may avoid or reduce the formation of cracks in the step of using the display device **60** to form a curved display device.

(84) Referring to FIG. 6B, a schematic side view of the display device **70** according to another embodiment of the present disclosure is provided. Similar to the display device **60**, the display device **70** also includes a portion **70a** corresponding to the curved portion of the curved display device (e.g., the curved portion **100** of the curved display device **10** illustrated in FIG. 1A) formed

of the display device **70**, and a portion **70b** corresponding to the non-curved portion of the curved display device formed of the display device **70**.

(85) In some embodiments, the dimension of one of the light-emitting units **206a** in the portion **70a** may be smaller than the dimension of one of the light-emitting units **206b** in the portion **70b**, which may avoid or reduce the formation of cracks in the step of using the display device **70** to form a curved display device. For example, the bottom surface area of one of the light-emitting units **206a** may be smaller than the bottom surface area of one of the light-emitting units **206b**. In some embodiments, the ratio of the bottom surface area of one of the light-emitting units **206a** to the bottom surface area of one of the light-emitting units **206b** may be 0.3 to 0.95.

(86) In some embodiments, as shown in FIG. **6B**, the density (e.g., number of light-emitting units per unit area) of the light-emitting units **206a** in portion **70a** may be smaller than the density of the light-emitting units **206b** in portion **70b**, which may avoid or reduce the formation of cracks in the step of using the display device **70** to form a curved display device.

(87) It should be noted that the curved display devices formed of the display devices of the above embodiments may include technical features which are the same as or similar to the technical features of the display devices of the above embodiments (e.g., the technical features of the light-emitting units discussed in paragraphs [0093]-[0094], and [0096]-[0097]). These curved display devices are intended to be included within the scope of the present disclosure.

Fifth Embodiment

(88) Fifth Embodiment of the present disclosure also provides some display devices which may avoid or reduce the formation of cracks in the step of using these display devices to form curved display devices.

(89) Referring to FIG. **7A**, a schematic top view of the display device **80** according to Fifth Embodiment of the present disclosure is provided. Although only the substrate **800** and the conductive electrodes (or conductive lines) **802** are shown in FIG. **7A** for the interest of simplicity, one skilled in the art can understand that additional layers and/or components discussed in the above embodiments may also be disposed in the display device **80**. For example, some light-emitting units or insulating layers may be formed in the display device **80**.

(90) For example, the display device **80** can be bent to form a curved display device (e.g., the curved display devices **10** or **10'** shown in FIGS. **1A** and **1B**). As shown in FIG. **7A**, the display device **80** can include a portion **80a** and a portion **80b**. For example, the portion **80a** can correspond to a curved portion of a curved display device (e.g., the curved portion **100'** of the curved display device **10'** illustrated in FIG. **1B**) formed of the display device **80**, and the portion **80b** can correspond to a non-curved portion of the curved display device formed of the display device **80**.

(91) As shown in FIG. **7A**, the conductive electrode (or conductive line) **802** can include a neck portion **802a** disposed in the portion **80a** of the display device **80**, and a portion **802b** disposed in the portion **80b** of the display device **80**. For example, the portion **802a** is narrower than the other portions of the conductive electrode (or conductive line) **802**. In some embodiments, a width $W_{sub.2}$ of the portion **802a** of the conductive electrode (or conductive line) **802** is smaller than a width $W_{sub.1}$ of the portion **802b** of the conductive electrode (or conductive line) **802**, which may avoid or reduce the formation of cracks in the step of using the display device **80** to form a curved display device.

(92) In some embodiments, as shown in FIG. **7B**, the conductive electrode (or conductive line) **802** can include one or more openings **804**. The density (e.g., number of openings per unit area) of the openings **804** in the portion **802a** of the conductive electrode (or conductive line) **802** and the density of the openings **804** in the portion **802b** of the conductive electrode (or conductive line) **802** may be different. For example, the density of the openings **804** in the portion **802a** of the conductive electrode (or conductive line) **802** may be greater than the density of the openings **804** in the portion **802b** of the conductive electrode (or conductive line) **802**, which may avoid or

reduce the formation of cracks in the step of using the display device **80** to form a curved display device.

(93) In some embodiments, as shown in FIG. 7C, the area of one of the openings **804** in the portion **802a** of the conductive electrode **802** and the area of one of the openings **804** in the portion **802b** of the conductive electrode (or conductive line) **802** may be different. For example, the area of one of the openings **804** in the portion **802a** of the conductive electrode (or conductive line) **802** may be greater than the area of one of the openings **804** in the portion **802b** of the conductive electrode (or conductive line) **802**, which may avoid or reduce the formation of cracks in the step of using the display device **80** to form a curved display device.

(94) It should be noted that the curved display devices formed of the display devices of the above embodiments may include technical features which are the same as or similar to the technical features of the display devices of the above embodiments (e.g., the technical features of the conductive electrodes discussed in the preceding three paragraphs. These curved display devices are intended to be included within the scope of the present disclosure.

(95) Understandably, a deposition process, a lithography process, an etching process, other applicable processes, or a combination thereof may be used to form the display devices of the above embodiments.

(96) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure. In addition, each claim can be an individual embodiment of the present disclosure, and the scope of the present disclosure includes the combinations of every claim and every embodiment of the present disclosure.

Claims

1. An electronic device, comprising: a substrate; a metal layer overlapped with the substrate and comprising a plurality of holes, wherein in a top view, an area of one of the holes is different from an area of another one of the holes; a plurality of diodes disposed on the substrate; and a first transistor and a second transistor disposed on the substrate and electrically connected to one of the plurality of diodes, wherein the first transistor has a first gate electrode, the second transistor has a second gate electrode, and the first gate electrode and the second gate electrode overlap a same semiconductor layer, wherein in a cross-sectional view, the one of the plurality of diodes overlaps the first transistor and does not overlap the second transistor, wherein the metal layer comprises a metal portion, the plurality of holes are separated by the metal portion, and one of the plurality of diodes does not overlap the metal portion.
2. The electronic device according to claim 1, wherein the electronic device is a bendable display.
3. The electronic device according to claim 2, wherein when the electronic device is bent to be a curved display, the one of the plurality of holes is located in a curved portion of the curved display, and the another one of the plurality of holes is located in a non-curved portion of the curved display.
4. The electronic device according to claim 3, wherein the area of one of the plurality of holes is greater than the area of the another one of the plurality of holes.
5. The electronic device according to claim 3, wherein one of the plurality of diodes in the curved portion of the curved display is smaller than another one of the plurality of diodes in the non-curved portion of the curved display.

6. The electronic device according to claim 1, further comprising a pixel defining layer adjacent to the one of the plurality of diodes.

7. The electronic device according to claim 6, wherein the pixel defining layer overlaps the second transistor.
